



MINOR PROJECT SYNOPSIS REPORT

For

AI – Based Tourism Pressure and Ecosystem Stress Prediction

Submitted By

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Introduction

Tourism is a major contributor to the economy of Uttarakhand, especially due to pilgrimage activities in the Char Dham region comprising Kedarnath, Badrinath, Gangotri, and Yamunotri. Every year, a large number of tourists visit these sites within a limited seasonal window, which creates significant pressure on the surrounding environment. The fragile Himalayan ecosystem is highly sensitive to factors such as deforestation, land-use change, waste generation, water consumption, and changing climate patterns.



In recent years, increasing tourist footfall combined with climate variability has raised concerns about ecosystem stress in the Char Dham region. During peak pilgrimage seasons, the sudden rise in human activity often exceeds the natural carrying capacity of these areas, leading to long-term environmental degradation. Understanding the relationship between tourism pressure and ecological stress is therefore essential for sustainable tourism management.



This project aims to develop an AI-based approach to analyze and predict ecosystem stress in the Char Dham region using historical tourism data, satellite imagery, and climate information. By applying time-series forecasting, regression analysis, and GIS-based feature extraction, the study seeks to identify critical tourism thresholds and provide early warning insights to support environmentally responsible tourism planning.

Technical Concepts

1. Data Collection & Preprocessing

- Secondary data collection (tourism, climate, satellite data)
- Data cleaning, normalization, and aggregation

2. Exploratory Data Analysis (EDA)

- Trend and seasonality analysis
- Correlation analysis between tourism, climate, and ecosystem variables

3. Time Series Forecasting

- ARIMA for seasonal tourist footfall forecasting
- LSTM for long-term tourism trend prediction

4. Regression & Statistical Analysis

- Multiple linear regression
- Impact analysis of tourism and climate factors on ecosystem stress

5. Remote Sensing & GIS Analysis

- Land Use and Land Cover (LULC) classification
- Forest cover change detection
- NDVI-based vegetation stress analysis

6. Feature Engineering & Importance Analysis

- Tourism pressure and ecosystem stress index formulation
- Feature importance evaluation

7. Correlation & Impact Assessment

- Tourism vs forest degradation analysis
- Tourism-induced vs climate-induced stress comparison
- Waste consumption correlation with tourist inflow

8. Model Evaluation & Visualization

- Model performance evaluation (RMSE, R^2)
- GIS-based visualization and dashboards
- Early warning indicators for ecosystem stress

Motivation

Being from India, we have personally seen how pilgrimage tourism affects hill regions, especially during peak seasons. Places like Kedarnath, Badrinath, Gangotri, and Yamunotri become extremely crowded during the Char Dham Yatra. Roads remain jam-packed for hours, waste is visible along pathways, and natural areas that were once clean and peaceful face heavy human pressure.

During and after the yatra season, issues such as overflowing garbage, water pollution, tree cutting for road expansion, and increased landslide risks become clearly noticeable. These problems are not something we only read about; they are visible on news channels, social media, and even during personal travel experiences. Despite this, most actions are taken only after damage has already occurred.

This motivated us to think about whether technology can help predict such pressure in advance. With the availability of tourism data, climate records, and satellite images, Artificial Intelligence can be used to identify patterns and provide early warnings. The motivation of this project is to create a data-driven system that helps manage tourism pressure responsibly, protects fragile ecosystems, and supports sustainable development of the Char Dham region.

Problem Statement

The Char Dham region of Uttarakhand is facing increasing environmental pressure due to a rapid rise in seasonal tourism and changing climate conditions. High tourist footfall during peak periods leads to vegetation loss, land-use changes, increased waste generation, and stress on water resources. However, there is no integrated, data-driven system to measure and predict how tourism and climate factors together impact the fragile ecosystem of this region. This lack of timely assessment makes it difficult to identify safe tourism limits and take preventive actions. Therefore, a predictive approach using tourism data, satellite imagery, and climate information is required to assess ecosystem stress and support sustainable tourism planning.

Areas of Application

Sustainable Tourism Planning

Helps estimate safe tourist limits and manage seasonal overcrowding in the Char Dham region. It will help people to get an idea of what situation there exists.

Early Warning System

Identifies periods of high ecosystem stress in advance, from which actions can be taken before any unusual occurrence.

Vegetation and Forest Monitoring

Tracks Forest cover loss and vegetation stress linked to tourism activities. And also, it will help in planning future routes and maps.

Waste and Water Management

Supports better planning of waste disposal and tracking of non-civic activities in these places.

Policy and Decision Support

Provides data-driven insights for sustainable tourism and environmental policies. This will help authorities to make justifiable actions and leadings.

Scalability

Can be adapted to other ecologically sensitive tourism regions like Himalayas, South-India Tourism, Kashmir, etc.

Data Sources

A. Tourist Footfall (Season / Month-wise)

- Uttarakhand Tourism Development Board
- District Tourism Offices
- Char Dham Yatra official reports

B. Satellite & GIS Data

- Landsat (USGS) – long-term forest change (2000–present)
- Sentinel-2 (ESA) – high-resolution land-use mapping

C. Climate Data

- NASA POWER Climate Dataset
- WorldClim

D. Waste Consumption

- Swachh Bharat Mission reports
- Research papers on pilgrimage waste

Literature Review

- Several studies on Himalayan pilgrimage areas have pointed out that a sudden rise in tourist numbers, especially during peak seasons, has led to forest loss, land-use changes, and growing environmental risks in these fragile regions. (*Brandt et al., 2019*)
- Researchers have widely used satellite data such as Landsat and MODIS to observe changes in vegetation and forest cover over time. Indices like NDVI have been effective in showing how natural vegetation in Uttarakhand and nearby Himalayan areas has gradually declined. (*ScienceDirect, 2021*)
- Studies conducted along popular pilgrimage and trekking routes show that waste generation and water stress increase sharply with tourist inflow, particularly during short pilgrimage seasons when local infrastructure is unable to cope with the load. (*ScienceDirect, 2003*)
- Time-series models like ARIMA and LSTM have been commonly used to forecast tourist arrivals and seasonal tourism demand. These models help in understanding long-term trends and peak pressure periods. (*ScienceDirect, 2018*)
- Most existing research looks at tourism growth, climate effects, and environmental change as separate problems. Very few studies combine tourism data, satellite-based ecosystem indicators, and climate variables in a single framework. (*ScienceDirect, 2009*)

Research Gap: There is a clear need for an integrated approach that uses AI and GIS together to study tourism pressure and ecosystem stress, particularly for sensitive pilgrimage regions like the Char Dham.

Objective

The main objective of this project is to develop an AI-based system that can analyze and predict tourism pressure and ecosystem stress in the Char Dham region using historical and real-world data. The system aims to support sustainable tourism while protecting the fragile natural environment.

Specific Objectives:

- To collect and analyze historical tourism, climate, satellite, and waste-related data for the Char Dham region.
- To identify key indicators that reflect environmental stress, such as vegetation loss and waste generation.
- To forecast tourist footfall during peak and off-season periods using time-series and machine learning models.
- To study the relationship between tourism activities and ecosystem degradation through data-driven analysis.
- To classify stress levels into risk categories (low, medium, high) for easier understanding.
- To develop a visual dashboard that presents insights and early warnings for effective decision-making by authorities.

Methodology

A) Reference Software Model

The Incremental Model is suitable for this project because the system is developed in well-defined stages and gradually enhanced over time. Initially, the project focuses on historical data analysis and model development, and later extends to real-time data integration and dashboard deployment.

Increment-wise Mapping:

- **Increment 1:** Data collection, preprocessing, and EDA
- **Increment 2:** Tourism forecasting (ARIMA/LSTM) and regression analysis
- **Increment 3:** Satellite & GIS-based ecosystem stress analysis
- **Increment 4:** Integration of results and dashboard visualization

B) Steps

Step 1: Data Collection

- Collect tourism, climate, waste datasets
- Data cleaning & normalization
- Exploratory Data Analysis (EDA)
- Satellite data sourcing
- GIS boundary mapping (Char Dham regions)
- Initial feature extraction

Step 2: GIS & Feature Engineering

- Forest cover change detection
- Land-use classification (LULC)
- NDVI-based ecosystem stress metrics
- Feature correlation analysis

Step 3: Modeling & Prediction

- ARIMA & LSTM for tourist footfall forecasting
- Climate–tourism interaction modeling
- Regression & feature importance analysis
- Ecosystem stress prediction model
- Threshold identification

Step 4: Dashboard & Early Warning System

- Interactive dashboard (Power BI / Streamlit)
- Risk alerts (High / Medium / Low stress)
- Visual maps & trend charts

C) Timeline

Week	Work
Week 1	Synopsis & planning
Week 2	Data collection
Week 3	Data cleaning
Week 4	EDA
Week 5	Modeling
Week 6	Stress analysis
Week 7	Dashboard
Week 8	Final report & PPT